

Miguel Ricardo
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Mexico

Miguel Ricardo González Márquez

Ph.D. in Sciences | Data Scientist & AI Architect

Summary Data Scientist and Ph.D. with over 4 years of experience building predictive models, optimizing machine learning architectures, and deploying end-to-end data products. Adept at handling structured and unstructured data using SQL, Python, and modern cloud architectures (Node.js, Vue.js). Educator, specializing in translating complex statistical methodologies into actionable business insights for non-technical stakeholders. Working in bridging the gap between rigorous experimental design and production-ready machine learning deployments.

Education

2007-2011, B.S. in Computer Systems Engineering

Instituto Tecnológico de Agua Prieta (TecNM/ITAP)

2016-2018, M.S. in Engineering

Universidad Autónoma de Baja California (UABC)

2018-2023, Ph.D. in Sciences

Universidad Autónoma de Baja California (UABC)

Professional Experience

March 2022 - Present, Lead AI Researcher & Data Scientist, Samsung Research Tijuana

Lead advanced R&D initiatives utilizing predictive modeling, clustering, and classification to solve complex pattern recognition problems. Responsible for designing rigorous statistical experiments, balancing bias-variance tradeoffs, and deploying robust data pipelines. Serve as a core technical mentor translating complex ML metrics into actionable insights and training engineering teams to establish a data-driven culture.

January 2015 - June 2016, IT Support & Network Technician, Instituto Tecnológico de Agua Prieta

Provided technical support for campus IT infrastructure; installed, maintained, and configured the campus network; developed and implemented institutional software projects utilizing relational databases.

Teaching & Mentorship Experience

August 2024 - Present, Adjunct Professor, CETYS Universidad Ensenada

Lecturer for Software Engineering and Digital Graphic Design Engineering programs. Fostering data literacy and a data-driven mindset among students by designing project-based curricula focused on modern full-stack architectures, API integrations, and scalable frontend frameworks.

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Communication & Leadership

2025, *Panelist*, 4th Colloquium on Applications of Artificial Intelligence in Science

Roundtable panelist discussing the integration of AI, predictive analytics, and scalable data systems in scientific research.

2025, *Speaker*, Samsung Week - INOVATEC - Instituto Tecnológico de Tijuana

Conference presentation: "Image Generation with Artificial Intelligence and the Importance of Robust Data Strategies."

2024, *Speaker*, INOVATEC - Instituto Tecnológico de Tijuana

Conference presentation: "The Necessity of Lightweight AI Models in Production Environments."

2024, *Speaker*, DIGEEK 2024 CETYS Ensenada

Keynote presentation: "New AI Technologies and Data Storytelling."

2020, *Oral Presentation*, ROPEC 2020 (Ixtapa, Guerrero)

Virtual presentation at the 22nd IEEE Autumn Meeting on Power, Electronics and Computing.

2016, *Instructor*, COMEST 2016

Delivered a comprehensive Web Development course at Instituto Tecnológico de Agua Prieta for Computer Systems Engineering students.

Technical Skills

Data Science & Analytics

- Python & R
- Scikit-Learn
- SQL
- Experimental Design
- Statistics

Machine Learning

- Predictive Modeling
- PyTorch & CUDA
- Neural Networks
- LangChain
- OpenCV

Production & Cloud Architecture

- Node.js & Vue 3
- REST APIs
- ML Model Deployment
- C++ / TensorRT
- JavaScript

Awards

2018, *Academic Merit Award*, UABC

Awarded for achieving the highest GPA in the Master's in Engineering program.

2011, *Valedictorian*, TecNM/ITAP

Awarded for achieving the highest GPA in the Computer Systems Engineering graduating class.

Interests

Professional

Passionate about bridging the gap between rigorous scientific experiments and production-ready business solutions. Focused on applied predictive modeling, translating complex data into actionable insights, and fostering a data-driven culture among technical and non-technical teams alike.

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Scientific Publications

Comparison of CNN architectures for single grape detection

<https://doi.org/10.1016/j.compag.2025.109930>

Multiple circle detection in images: a simple evolutionary algorithm approach and a new benchmark of images

<https://doi.org/10.1007/s10044-021-01007-6>

Grape Bunch Detection Using A Pixel-Wise Classification In Image Processing - IEEE Conference Publication

<https://doi.org/10.1109/ROPEC50909.2020.9258707>

Method to extract an enhanced cervical vertebrae area from a digital X-ray image

<https://doi.org/10.1016/j.mex.2018.06.009>

Professional References

Dr. Carlos Alberto Brizuela Rodríguez

Researcher, Department of Computer Science, Ensenada Center for Scientific Research and Higher Education (CICESE), Ensenada, B.C.
Email: cbrizuel@cicese.edu.mx

Dr. Miguel Enrique Martínez Rosas

Professor Emeritus, Faculty of Engineering, Architecture and Design, UABC, Ensenada Campus.
Email: emartine@uabc.edu.mx

Master Elizabeth Santillán Tarazón

Head of the IT Department, Instituto Tecnológico de Agua Prieta (TecNM/ITAP), Agua Prieta, Sonora.
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Master Ana Silvia Armenta Martínez

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